A Spatial Decision Support System for Sustainable and Climate-Resilient Data Centre Site Selection in Germany

MSc Thesis Research Proposal

Emmanuel Frimpong Kissi

s59ekiss@uni-bonn.de | University of Bonn

Supervisor: Prof. Matthias Garschagen; Dr-Ing Mahsa Moghadas

1 Background and Research Problem

Data centres have become the backbone of the digital economy, supporting a wide range of services, including cloud computing and artificial intelligence. Their rapid expansion is increasing demand for data-centre infrastructure while raising concerns about energy, water and land requirements (Al Kez et al., 2022). The location of a data centre can influence its energy consumption, environmental impacts, resource requirements, operating conditions and exposure to relevant risks (Daim et al., 2013; Depoorter et al., 2015). Data-centre siting therefore extends beyond conventional infrastructure and economic considerations to include sustainability, climate resilience and spatial planning.

Sustainability is important because location affects the long-term environmental performance and resource requirements of data-centre operations. European research shows that location influences cooling demand, opportunities for free cooling, renewable-energy potential and the environmental characteristics of electricity supply (Depoorter et al., 2015). Sustainable siting therefore requires consideration of energy and carbon performance, renewable-energy availability, water-resource sustainability and ecological effects. These considerations go beyond infrastructure availability, since a well-connected location may still have unfavourable environmental or resource conditions (Erdem & Özdemir, 2024; Kheybari et al., 2020).

Climate resilience represents a distinct but complementary requirement. Extreme temperatures can increase cooling demand, while flooding, drought and water stress can increase operational risks and resource constraints. In Germany, Turek and Radgen (2021) demonstrated substantial spatial variation in data-centre cooling-energy consumption associated with mesoclimatic conditions, highlighting the importance of climate conditions in location decisions.

In addition, infrastructure provision, resource availability and spatial and planning compatibility represent distinct components of the siting decision. Infrastructure provision concerns access to electricity, telecommunications, transport and utility networks; resource availability concerns the longer-term availability and sustainability of resources such as energy and water; while spatial and planning compatibility concerns land use, protected areas, environmental restrictions and regulatory

requirements. Recent studies have integrated infrastructure, accessibility, hazards and land availability within spatial data-centre assessments (Arzumanyan et al., 2025; Lee & Rusli, 2025). GIS-based research has also demonstrated the importance of distinguishing weighted suitability factors from areas that should be excluded because of environmental or planning constraints (Hussain et al., 2026).

2 Research Gap

Existing data-centre siting studies consider factors such as infrastructure, energy, environmental conditions, climate risks, accessibility and land-use constraints, but these are addressed through different frameworks and geographic contexts (Arzumanyan et al., 2025; Daim et al., 2013; Hussain et al., 2026; Kheybari et al., 2020; Lee & Rusli, 2025; Turek & Radgen, 2021). The research gap lies in the lack of a coherent framework that systematically integrates sustainability, climate resilience, infrastructure, resource availability and spatial and planning compatibility for data-centre siting in Germany. This study addresses the gap by synthesising these requirements, validating them with experts, spatially operationalising them through a spatial decision support system using GIS-MCDA, and testing the robustness of the resulting suitability patterns.

3 Aim and Objectives

3.1 Aim

To develop and spatially operationalise an expert-informed spatial decision support system for identifying suitable locations for sustainable and climate-resilient data-centre development in Germany.

3.2 Objectives

1. Synthesize the literature to develop a preliminary hierarchy of dimensions, criteria and spatial indicators for sustainable and climate-resilient data-centre siting.
2. Validate and refine the preliminary framework through expert assessment, identifying relevant, missing or redundant criteria and distinguishing weighted factors from exclusion constraints.
3. Determine subjective criterion weights using an appropriate expert-based MCDA method, with the Best-Worst Method (BWM) considered as a candidate approach.
4. Spatially operationalise the validated framework using appropriate datasets for Germany within a GIS-MCDA environment.
5. Assess the sensitivity and robustness of the resulting suitability patterns to plausible variations in expert-derived criterion weights.

4 Research Questions

1. Which dimensions, criteria and spatial indicators are relevant for assessing sustainable and climate-resilient data-centre siting in Germany?

2. How do experts evaluate the relevance and relative importance of the identified criteria, and how can their assessment be used to validate and refine the conceptual framework?
3. Where are relatively highly suitable areas for sustainable and climate-resilient data-centre development in Germany when the validated framework is implemented through GIS-MCDA?
4. How robust are the resulting spatial suitability patterns to plausible variations in expert-derived criterion weights?

5 Proposed Methodology

The study will follow a sequential workflow of literature synthesis, conceptual framework development, expert validation, subjective weighting, GIS-MCDA, and sensitivity and robustness assessment. Peer-reviewed literature on data-centre siting, sustainability, climate resilience, infrastructure, resource availability, and spatial planning will first be synthesized to establish a hierarchy of dimensions, criteria, and spatial indicators. Criteria will be selected according to conceptual relevance and empirical support rather than data availability alone.

The preliminary framework will be evaluated by experts in data-centre infrastructure, energy, environmental management, climate risk, and spatial planning. Expert assessment will identify missing, redundant, or poorly defined criteria and distinguish weighted suitability factors from hard exclusion constraints. Following refinement, subjective criterion weights will be derived using an appropriate expert-based MCDA method, with the Best-Worst Method (BWM) considered as a candidate approach (Rezaei, 2015). Validated criteria will subsequently be operationalised spatially using suitable datasets for Germany, harmonised within a common GIS framework, and standardised according to theoretically justified suitability relationships. Weighted factors will be integrated through GIS-MCDA, while legally or physically unsuitable areas will be excluded separately.

Finally, sensitivity analysis will test plausible variations in expert-derived weights and weighting scenarios to assess the robustness of the resulting suitability patterns.

6 Expected Contribution

The study will provide a literature-grounded and expert-informed spatial decision-support framework for sustainable and climate-resilient data-centre siting in Germany. Its contribution lies in systematically integrating diverse siting requirements into a coherent spatial framework, translating them into measurable criteria and indicators, distinguishing weighted suitability factors from exclusion constraints, and evaluating the robustness of suitability results to variations in expert-derived weights.

References

1. Al Kez, D., Foley, A. M., Laverty, D., Del Rio, D. F., & Sovacool, B. (2022). Exploring the sustainability challenges facing digitalization and internet data centers. *Journal of Cleaner Production, 371*, 133633. https://doi.org/10.1016/j.jclepro.2022.133633
2. Arzumanyan, M., Calzado, E. R., Lin, N., Bahadur, V., Das, J., Ko, T. L., & Koesterke, L. (2025). Geospatial suitability analysis for data center placement: A case study in Texas, USA. *Sustainable Cities and Society, 131*, 106687. https://doi.org/10.1016/j.scs.2025.106687
3. Daim, T. U., Bhatla, A., & Mansour, M. (2013). Site selection for a data centre – a multi-criteria decision-making model. *International Journal of Sustainable Engineering, 6*(1), 10–22. https://doi.org/10.1080/19397038.2012.719554
4. Depoorter, V., Oró, E., & Salom, J. (2015). The location as an energy efficiency and renewable energy supply measure for data centres in Europe. *Applied Energy, 140*, 338–349. https://doi.org/10.1016/j.apenergy.2014.11.067
5. Erdem, M., & Özdemir, A. (2024). Sustainability and risk assessment of data center locations under a fuzzy environment. *Journal of Cleaner Production, 450*, 141982. https://doi.org/10.1016/j.jclepro.2024.141982
6. Hussain, S. N., Al-Mandhari, M., Ali, S. M. F., Zaib, A., & Ghosh, A. (2026). GIS-driven regional assessment for sustainable data center siting in the United Kingdom. *Land, 15*(3), 516. https://doi.org/10.3390/land15030516
7. Kheybari, S., Davoodi Monfared, M., Farazmand, H., & Rezaei, J. (2020). Sustainable location selection of data centers: Developing a multi-criteria set-covering decision-making methodology. *International Journal of Information Technology & Decision Making, 19*(3), 741–773. https://doi.org/10.1142/S0219622020500157
8. Lee, M. Z., & Rusli, N. (2025). A multicriteria decision-making approach for identifying the suitable data center location in Iskandar Puteri, Johor. *Chinese Journal of Urban and Environmental Studies, 13*(4), 2550021. https://doi.org/10.1142/S2345748125500216
9. Rezaei, J. (2015). Best-worst multi-criteria decision-making method. *Omega, 53*, 49–57. https://doi.org/10.1016/j.omega.2014.11.009
10. Turek, D., & Radgen, P. (2021). Optimized data center site selection—mesoclimatic effects on data center energy consumption and costs. *Energy Efficiency, 14*(3), 33. https://doi.org/10.1007/s12053-021-09947-y